Research Article

Supplement to Induced Paths, Radius, and Extremal Neighbourhood Independence in Graphs without 4-Cycles

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Abstract

This supplement contains the rooted bounds, the endpoint obstructions, the high-radius constructions, the candidate pairs in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$, the one-triangle bound, the edge-deficit estimates, and the finite-search certificate for Lemma 5.7. Numbered statements carry an S prefix. The dependency note below is the only list of which arguments are hypotheses of the article. The certificate used by Theorem 5.8 is the file `certificates/gen3/verify_gen3.py` at commit `dd555d305b4a69076cafb845087f00b4144f07cc`, run from the repository root by `python3 certificates/gen3/verify_gen3.py`. Appendix A (Section A) records an independent execution of that command.

DEPENDENCY ON THE ARTICLE

Theorems 3.6, 4.3, 5.5 and 5.6 of the article do not use this file. Theorem 5.8 uses only Section A, through Lemma 5.7, together with Lemma 5.3 proved in the article. Theorem 5.9 uses Theorem 5.8. The rows below separate that certificate from the material that is not a hypothesis of those statements.

Sections 2–5 keep the class restrictions of the statements they prove. Outside those statements, the following are open and are not hypotheses of the article: ROOT(q) for $q \geq 5$ on an arbitrary strong C_4-free graph, GEN_K for $K \geq 4$, RAD(r) for $r \geq 5$, and a structure theorem for radius $r \geq 5$ and path order $2r - 1$. The candidate pairs in Section 4 are necessary conditions only. The conjectures there are not proofs of REAL-133. Example S3.5 does not fix a density cutoff or count eligible cores, and Example S3.6 is a scalar system that does not imply the conjecture. The graph $K_{2,3}$ contains a 4-cycle, so it does not test the block restriction inside the strong C_4-free class.

Material	Role
Section A	used by Lemma 5.7 and Theorem 5.8
Sections 1, 2, 3 and 4	not used by the article
Theorem S5.1 and Lemmas S5.2, S5.3	alternate route for one and two triangles; not used by Theorem 5.8
Lemmas S5.5, S5.6 and S5.7	edge-deficit estimates; not used by Theorem 5.8
Conjectures S4.4, S4.5 and S4.6	open, and not hypotheses of any proved statement

1 | Rooted bounds of orders three and four

Theorem S5.1 uses Lemma 5.3 from the article. The rooted bound that bears on the next open band is Theorem S1.2: it is the estimate which shows that a minimum counterexample to GEN_4 is 2-connected. The distance-layer form of the same estimate is omitted. The block-class bound and the Moore observation are kept because they delimit GEN_K, and Example S3.4 is kept because it is the explicit failure of the endpoint inequalities at order nine.
For a vertex s of a connected graph F let $p_s(F)$ be the largest order of an induced path starting at s. Write $D = d_F(s)$ and $m = e(F[N(s)])$.

Definition S1.1. For an integer $q \geq 3$, the rooted bound ROOT(q) asserts that every connected strong C_4-free graph F and every vertex s with $p_s(F) \leq q$ satisfy

$$S(F) \leq (q-1)(|F|-1) - (q-3)D - m.$$

The hypothesis $p_s(F) \leq q$ belongs in the statement. Without it the displayed inequality is false: if $q = 3$ and $F = C_5$, then $S(F) = 10$ and $m = 0$, while $2(|F|-1) = 8$. Lemma 5.3, already proved because the cut-vertex induction of Theorem 5.5 uses it, is the case $q = 3$.

Theorem S1.2. *Let F be connected and strong C_4-free, and let $s \in V(F)$ satisfy $p_s(F) \leq 4$. Write $D = d(s)$ and $m = e(F[N(s)])$. Then*

$$S(F) \leq 3(n-1) - D - m. \tag{1.1}$$

In particular ROOT(4) *holds. Equality holds for C_5 at any root, and for a graph formed by $k \geq 1$ copies of C_5 that meet only at the root.*

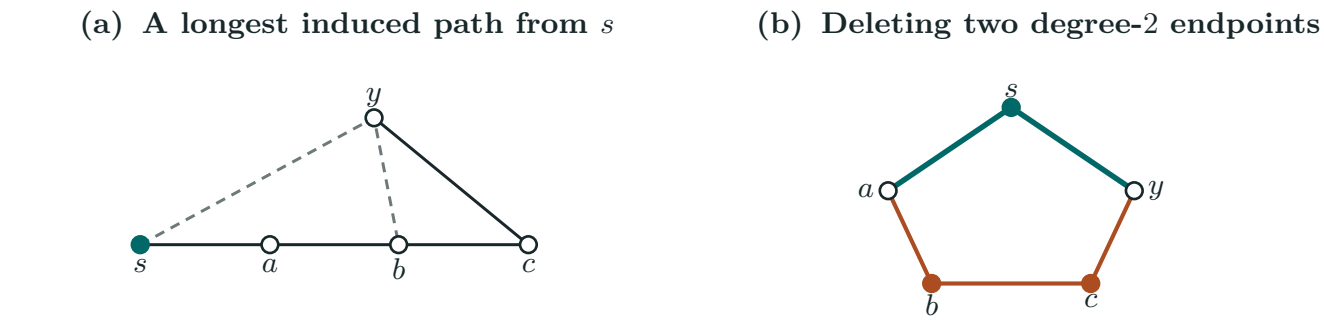


FIGURE 1 | Deletion at the end of a longest induced path of order four starting at s. (a) Each neighbour $y \neq b$ of c must be adjacent to s or b. Dashed lines indicate alternatives. (b) If neither terminal vertex c nor b can be deleted with loss at most three, the indicated 5-cycle is induced and both have degree two. The deletion of $\{b, c\}$ has loss six and preserves connectivity.

Proof. The proof is by induction on n. If $p_s(F) \leq 3$, Lemma 5.3 gives $S \leq 2(n-1) - m$. Since $D \leq n-1$, the right-hand side is at most $3(n-1) - D - m$.
Now suppose $p_s(F) = 4$, and let s, a, b, c be an induced path. The endpoint c is not a cut vertex: a neighbour of c in a component of $F - c$ that does not contain s would extend the path. Let $y \in N(c) \setminus \{b\}$. The path s, a, b, c, y is not induced, so y meets at least one of s, a, b. The edge ya would give the 4-cycle a, b, c, y. Thus y is adjacent to s or to b. The vertices s and c have at most one common neighbour, and so do b and c, hence $d(c) \leq 3$. If $d(c) = 3$, then c has two neighbours besides b, and one of them is adjacent to b. So c lies in a triangle, $t(c) \geq 1$ and $L(c) = 2d(c) - 3t(c) \leq 3$. Figure 1 shows the two deletions.
If $L(c) \leq 3$, delete c. The graph $F - c$ is connected, and $c \notin N[s]$, so D and m are unchanged. The inductive hypothesis gives $S(F - c) \leq 3(n-2) - D - m$, and therefore $S(F) \leq 3(n-2) - D - m + 3 = 3(n-1) - D - m$.
The remaining case is $d(c) = 2$ and $t(c) = 0$, so $L(c) = 4$. Let y be the neighbour of c other than b. Then y is not adjacent to b, hence ys is an edge, and s, a, b, c, y is an induced 5-cycle: every chord is excluded by the induced path s, a, b, c, by $t(c) = 0$, or by the 4-cycle that ay would create. The vertex b is the endpoint of the induced path s, y, c, b. If $L(b) \leq 3$, delete b and apply the inductive hypothesis as above.

Otherwise the same endpoint analysis gives $d(b) = 2$ and $t(b) = 0$. Delete $U = \{b, c\}$. A path that used the edges ab, bc, cy can be replaced along a, s, y, so $F - U$ is connected. The deleted edges are ab, bc and cy, and no triangle is deleted. Since $S = 2e - 3T$, one has $D_F(U) = 6 = 3|U|$. The vertices b and c lie outside $N[s]$, so D and m are unchanged. Induction gives $S(F - U) \le 3(n - 3) - D - m$, and $S(F) \le 3(n - 3) - D - m + 6 = 3(n - 1) - D - m$.
For the equality examples, root C_5 at any vertex. Then $n = 5$, $D = 2$, $m = 0$ and $S = 10 = 3(n - 1) - D - m$. If k copies of C_5 meet only at the root, then $n = 1 + 4k$, $D = 2k$ and $m = 0$, because the two neighbours of the root on each copy are nonadjacent, and $S = 10k = 3(n - 1) - D - m$. □

Proposition S1.3. *Let F be connected and strong C_4-free, and let s satisfy $p_s(F) = 4$ and $\mathrm{ecc}(s) = 3$. Then $S(F) \le 3(n - 1) - D - m - 2$. Equality holds for P_4 rooted at an endpoint.*

Proof. The proof is by induction on n for instances with $p_s = 4$ and $\mathrm{ecc}(s) = 3$. Fix a vertex z at distance 3 from s. Whenever an endpoint other than z exists, the deletion in the proof of Theorem S1.2 applies to it. In the exceptional 5-cycle the vertices b and c are at distance 2 from s, so the deletion leaves z. The remaining graph is connected and strong C_4-free, D and m are unchanged, and distances do not decrease. Thus $d(s, z) \ge 3$. Since $p_s \le 4$, no distance exceeds 3, so $p_s = 4$ and the eccentricity of s is still 3. A deletion of one vertex of loss at most 3 contributes 3, and the inductive bound on $n - 1$ vertices contributes $3(n-2) - D - m - 2$, which sum to $3(n-1) - D - m - 2$. Deleting $\{b, c\}$ contributes 6 against $3(n - 3) - D - m - 2$.
If z is the unique endpoint, delete z. Then $p_s(F - z) \le 3$. An extra neighbour of z is not adjacent to s, for that would give $d(s, z) \le 2$. By the endpoint analysis it is adjacent to the predecessor of z and forms a triangle, so $L(z) \le 2$. Lemma 5.3 gives $S(F - z) \le 2(n - 2) - m$, hence $S(F) \le 2(n - 1) - m$. A geodesic from s to z has at least two vertices outside $N[s]$, so $n - 1 - D \ge 2$ and $2(n - 1) - m \le 3(n - 1) - D - m - 2$.
For P_4 rooted at an endpoint one has $n = 4$, $D = 1$, $m = 0$ and $S = 6$, and the strengthened bound is $3 \cdot 3 - 1 - 2 = 6$. □

Proposition S1.4. *Let G be a counterexample to* GEN_4 *of minimum order. Then G is 2-connected. This does not decide* GEN_4.

Proof. Such a graph is connected and strong C_4-free, with $p(G) \le 8$ and $S(G) \ge 4n + 1$. Let v be a cut vertex, and let G_1, G_2 be the graphs of Lemma 2.3. Write ρ_i for the largest order of an induced path of G_i that starts at v. Paths from the two sides form an induced path of G on $\rho_1 + \rho_2 - 1 \le 8$ vertices, so some side, say G_1, has $\rho_1 \le 4$. Theorem S1.2 gives $S(G_1) \le 3(n_1 - 1)$. The graph G_2 is a smaller connected strong C_4-free graph of path order at most 8, so minimality gives $S(G_2) \le 4n_2$. Therefore $S(G) \le 3(n_1 - 1) + 4n_2$. Since $n = n_1 + n_2 - 1$,

$$3(n_1 - 1) + 4n_2 = 4n - n_1 + 1.$$

The hypothesis $n_1 \ge 2$ gives $4n - n_1 + 1 \le 4n - 1$, so $S(G) < 4n$, contradicting $S(G) \ge 4n + 1$. □

For $q \ge 5$, $\mathrm{ROOT}(q)$ on an arbitrary strong C_4-free graph is open. The next section proves every index for graphs whose blocks are bipartite or cycles. Conjecture S4.5 implies $\mathrm{ROOT}(q)$ by the attachment in Section 4.

2 | Rooted bounds for every index

Let $\mathcal{B}$ be the class of connected strong C_4-free graphs in which every block is bipartite or a cycle. Edges are blocks, and cycles may be odd. The class contains every connected strong C_4-free bipartite graph and every connected strong C_4-free cactus. Theorem S2.2 keeps this block hypothesis. The graph $K_{2,3}$ contains a 4-cycle subgraph. Rooted in its part of size two it has $p_s = 3$ and $S = 12 > 2(n - 1) - m = 8$, so $\mathrm{ROOT}(3)$ fails once strong C_4-freeness is dropped.
A rooted graph (G, s) with $p_s(G) = q$ and $\mathrm{ecc}(s) = q-1$ is called *terminal*. For a terminal instance put $R = n-q+1-D \ge 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor (q - 1)/2 \rfloor$ for $q \ge 5$.

Lemma S2.1. *Let G be connected and strong C_4-free, and let P be a longest induced path starting at a vertex s, of order q. Let x be the other end of P, and suppose that the block B containing x is bipartite. Let Q be the subpath of P formed by the vertices of P that lie in B, and set $\ell = |Q|$. Then every neighbour of x lies in B, and x lies in no triangle. If $\ell = 2$, then $d(x) = 1$. If $\ell \ge 3$, then $d(x) \le \lfloor (\ell - 1)/2 \rfloor$. In every case $\ell \le q$, and $L(x) \le c_q$ whenever $q \ge 5$.*

Proof. A neighbour of x in a component of $G - x$ that does not contain s would extend P, so x is not a cut vertex. Every edge with one end in B and one end outside B is incident with a cut vertex of G belonging to B. Therefore every neighbour of x lies in B. The block is bipartite, so x lies in no triangle.
Since the block-cut tree is a tree, P enters B only once. The intersection of P with B is a single subpath $Q = z_0, \dots, z_{\ell-1}$ with $z_{\ell-1} = x$. If s lies in B, then $Q = P$. If s lies outside B, the vertices of P that precede z_0 lie outside B, and z_0 is their only

neighbour in B: any other edge from that prefix into B would be a second entry. A vertex of $B \setminus \{z_0\}$ is therefore adjacent to no vertex of the prefix. An extension of P at x through a vertex of B can fail only by a chord into Q.

Let y be a neighbour of x other than $z_{\ell-2}$. Then y lies in B, and y does not lie on Q, because Q is induced. The vertex y is not adjacent to the prefix. If y had no neighbour on $Q \setminus \{x\}$, the path P followed by y would be induced. Thus y has a neighbour z_i on $Q \setminus \{x\}$. That neighbour is not $z_{\ell-2}$, for $yxz_{\ell-2}$ would be a triangle, and it is not $z_{\ell-3}$, for $z_{\ell-3}z_{\ell-2}xy$ would be a 4-cycle. Since B is bipartite, z_i has the colour of x, so $i \equiv \ell - 1 \pmod 2$ and $i \le \ell - 5$.

If $\ell = 2$, then Q is a single edge and there is no admissible index, so $d(x) = 1$. If $\ell \ge 3$, the number of admissible indices is $\lfloor(\ell-1)/2\rfloor - 1$. Two neighbours of x are nonadjacent and cannot share a blocker, for a common blocker would form a 4-cycle with them and with x. Hence $d(x) \le \lfloor(\ell-1)/2\rfloor$. In every case $\ell \le q$. For $q \ge 5$ one has $d(x) \le \lfloor(q-1)/2\rfloor$, and $L(x) = 2d(x) \le c_q$.

The same count applies inside B. Let z be the cut vertex of G belonging to B and lying on every path from a fixed root to B, or the root itself if the root lies in B, and let Q be a longest induced path of B starting at z, with other end x and with $\ell = |Q|$. A neighbour of x in B that had no neighbour on $Q \setminus \{x\}$ would extend Q inside B. If the root lies outside B, no vertex of $B \setminus \{z\}$ meets the component of the root at z, so that component contributes no further blocker. Thus $d_B(x) \le 1$ if $\ell = 2$, and $d_B(x) \le \lfloor(\ell-1)/2\rfloor$ if $\ell \ge 3$. □

Theorem S2.2. *1. Every terminal instance in $\mathcal{B}$ satisfies $S(G) \le 2(D+q-2) - m + c_q R$.*

2. Every $F \in \mathcal{B}$ with a vertex s such that $p_s(F) \le q$ satisfies

$$S(F) \le c_q(n-1) - (c_q - 2)D - m.$$

In particular ROOT(q) *holds in $\mathcal{B}$ for every $q \ge 3$. If every block is bipartite or a triangle, one may take $c_q = 2\lfloor(q-1)/2\rfloor$ for all $q \ge 3$.*

Proof. Two terminal facts. An endpoint x of a longest rooted induced path is not a cut vertex: a neighbour of x in another component of $G - x$ would extend the path. If z is at distance $q-1$ from s and $v_0 = s, \dots, v_{q-1} = z$ is a geodesic, any neighbour $y \ne v_{q-2}$ of z must meet an earlier path vertex (else the rooted path extends); meeting v_i with $i \le q-4$ shortens $d(s,z)$, and meeting v_{q-3} gives a C_4; so $y \sim v_{q-2}$, and two such y would give a C_4. Hence $L(z) = 2d(z) - 3t(z) \le 2$. Moreover, if $x \ne s, z$, $G - x$ is connected and z survives, then $q - 1 \le d_{G-x}(s,z) \le p_s(G-x) - 1 \le q - 1$, so the instance stays terminal with the same q.

Endpoint losses. Let x be the endpoint of a longest rooted path of order q. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, and if a cycle of length at least 5 it is 4. If the block is bipartite, Lemma S2.1 gives $L(x) \le c_q$ for $q \ge 5$.

No nontrivial bipartite block when $q \le 4$. Let B be a nontrivial bipartite block, and let z be the cut vertex of G belonging to B that lies on every path from s to B, or let $z = s$ if $s \in B$. Let Q be a longest induced path of B starting at z, and let x be its other end. Every vertex of a nontrivial block has degree at least two in the block, so $d_B(x) \ge 2$. The last paragraph of Lemma S2.1 gives $d_B(x) \le 1$ if $|Q| \le 4$ and $d_B(x) \le \lfloor(|Q|-1)/2\rfloor$ in any case with $|Q| \ge 3$. The inequality $d_B(x) \ge 2$ therefore forces $|Q| \ge 5$. An induced s–z path meets B only at z. Concatenating it with Q yields an induced path that starts at s. The order of this path is $|Q|$ if $s \in B$, and at least $|Q|+1$ if $s \notin B$. In either case $p_s(G) \ge 5$. A terminal instance with $q \le 4$ therefore has no nontrivial bipartite block. Every block is an edge, a triangle, or a cycle of length at least five, so the graph is a cactus.

Bases. For $q \le 4$ the preceding paragraph leaves only cacti. For a C_4-free cactus with c_3 triangle blocks and c_L longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \le c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a C_5 through s. A chosen length-3 geodesic uses at most one edge of each such C_5: if it used two edges and then left the cycle, replacing that two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each C_5 contributes two vertices to R, $R \ge 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \ge 0$. For $q = 2$ the root is universal and $S = 2D - m$.

Induction on n. Let E be the set of endpoints of rooted induced paths of order q and z a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, x is not a cut vertex, its loss is at most c_q, the instance remains terminal with the same q, D and m, and R drops by one. If $E = \{z\}$, delete z: the new rooted maximum is $q-1$, the eccentricity $q-2$, R is unchanged and the loss is at most 2; induction gives $S \le 2(D+q-3) - m + c_{q-1}R + 2 \le 2(D+q-2) - m + c_q R$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.

(2) Attach a new path with $q-1$ edges at s. The result is terminal of index q, lies in $\mathcal{B}$, has order $n+q-1$, mass $S + 2(q-1)$ and root degree $D+1$, and m is unchanged. Substituting into (1) gives (2). Since $c_q \le q-1$ and $n - 1 - D \ge 0$,

$$\begin{aligned} c_q(n-1) - (c_q-2)D - m &= c_q(n-1-D) + 2D - m \\ &\le (q-1)(n-1-D) + 2D - m \end{aligned}$$

$$= (q-1)(n-1) - (q-3)D - m,$$

so ROOT(q) follows. □

A matching upper bound of shape $S \le (r-1)n - 2r(r-3)$, for graphs in $\mathcal{B}$ of radius $r \ge 5$ and path order $2r-1$, has been proposed from a structural description of radius-extremal graphs. That description is not proved in this paper, and the radius bound in $\mathcal{B}$ is not claimed.

Lemma S2.3. *Let G be strong C_4-free and triangle-free, and let $x_0x_1\cdots x_k$ be an induced path that cannot be extended at x_k. If $d(x_k) \ge 2$, then $k \ge d(x_k)+1$. If moreover G is bipartite, then $k \ge 2d(x_k)$.*

Proof. Let $y \ne x_{k-1}$ be a neighbour of x_k. It is not on the path, and, since the path cannot be extended, y has a neighbour x_i with $i \le k-1$. Here $i \ne k-1$ (a triangle) and $i \ne k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_ky$), so $i \le k-3$. A vertex x_i is adjacent to at most one neighbour of x_k. Hence the $d(x_k)-1 \ge 1$ such neighbours need distinct blockers among $x_0, \dots, x_{k-3}$, and $k-2 \ge d(x_k)-1$. If G is bipartite, the blocker has the colour of x_k, so $i \equiv k \pmod 2$ and $i \le k-4$; there are $\lfloor k/2 \rfloor - 1$ such positions, and $\lfloor k/2 \rfloor \ge d(x_k)$. □

Proposition S2.4. *Every connected regular bipartite strong C_4-free graph satisfies* GEN_K *for every K and satisfies REAL-133. Every Moore graph of diameter* 2 *and girth* 5 *satisfies REAL-133.*

Proof. Let G be d-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \ge 2$. As G is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma S2.3 applied to a longest induced path gives $p \ge 2d+1$, and Theorem 2.4 gives $p \ge 2r-1$; so $p \ge r+d$ both when $r \le d+1$ and when $r \ge d+1$. If $p \le 2K$, then $2d+1 \le 2K$, so $S = dn < Kn$. A Moore graph of diameter 2 and girth 5 has degree $d \ge 2$, radius 2 and $\ell = d$. The degree is an integer, so the floor form and the real form ask for the same inequality. The first part of Lemma S2.3 gives $p \ge d+2 = r+\ell$. This is an alternate proof of the case already recorded by Chen [1] immediately after his Theorem 3.6. It is not a new extremal classification. □

Remark S2.5. A Moore graph of degree 57, if it exists, has $n = 57^2+1 = 3250$ and $S = 57n$. Proposition S2.4 gives REAL-133, which only needs $p \ge 59$, and this conclusion is the integer case of the fact cited above. The statement GEN_{57} holds for it with equality in the mass bound, but GEN_{56} asks for $p \ge 113$ because $S > 56n$. We do not know whether such a graph would have $p \le 112$. If it did, it would refute GEN_{56} while satisfying REAL-133. Its existence would therefore not by itself separate the two statements, but it shows that a proof of every GEN_K is a different assertion from REAL-133.

3 | Negative results

The examples in this section are explicit graphs.

Example S3.1. Let $s \ge 3$, $t \ge 3$ and $m \ge 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of t edges at every cycle vertex, together with m internally disjoint paths c_0xzyc_1. Then $n = 2s(t+1)+3m$, $e = 2s(t+1)+4m$ and $T = 0$, so $e - n - T = m \ge 1$. For $(s,t,m) = (3,3,1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. They also show that $e = n+T$ fails for radius-extremal strong C_4-free graphs of every radius $r \ge 6$: here $r(G_{s,t,m}) = s+t$ and $p(G_{s,t,m}) = 2(s+t)-1$.
Two vertices have at most one common neighbour, so G is strong C_4-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and c_0 remains within distance two of every vertex of such a path, so the eccentricity of c_0 is still $s+t$. A vertex w of an added path meets the tip of the arm at c_s only through c_0 or c_1, hence at distance at least $s+t$. Thus $r = s+t$. For the path order, let B be the block formed by C and the added paths. An induced path P meets B in a subpath Q and can leave B only at an end of Q that is a cycle vertex, through at most t arm vertices. The same case division as for a single ear shows $|P| \le 2s+2t-1$, using $t \ge 3$ and $s \ge 3$, and the cycle minus one vertex together with the arms at its two ends attains this order.

The last sentence of Example S3.1 abbreviates a case division which we now record. If Q avoids both c_0 and c_1, then $|Q| \le 2s-2$. If Q contains exactly one of them, then either Q lies on the cycle with $|Q| \le 2s-1$, or $|Q| \le 2s+2$ with at most one cycle end, or $|Q| \le 7$ with no cycle end. If Q contains both, the same three shapes occur with $|Q| \le 2s-1$, $|Q| \le 2s+2$ and $|Q| \le 6$. Adding at most t arm vertices at each cycle end of Q gives $|P| \le \max\{2s-1+2t,\ 2s+2+t,\ 7\} = 2s+2t-1$.

Example S3.2. Forbidding induced paths on eleven vertices and requiring $L(v) \ge 6$ at every vertex does not forbid an induced star $K_{1,6}$. Two Petersen graphs sharing a vertex give a triangle-free strong C_4-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, $p = 9$ and $\lambda = 6$ at the shared vertex. For $b \ge 2$, gluing b copies of the Petersen graph along a common edge gives a 2-connected graph B_b with $n = 8b+2$, $S = 28b+2$, minimum loss 6, $p = 9$ and $\lambda = 2b+1$

at the two hubs. Thus singleton losses do not bound λ. These graphs are not counterexamples to GEN_5: the eight non-hub vertices of one Petersen block have deletion loss $28 < 5 \cdot 8 + 1$.

For an oriented edge (u, v) of a connected strong C_4-free graph G, let q be the maximum order of an induced path whose first two vertices are u, v in that order, and let E be the set of terminal vertices of the induced paths of that order. The endpoint inequality $\mathrm{EP}(q)$ asserts $D_G(E) \le q|E|$ for every such oriented edge of maximum order q. For a root s with $p_s(G) = q$, if E is the set of terminal vertices of the induced paths of order q starting at s, the rooted inequality $\mathrm{EP}^+(q)$ asserts $D_G(E) \le (q-1)|E|$.

Proposition S3.3. *For every integer $q \ge 4$, $\mathrm{EP}(q)$ holds for all connected strong C_4-free graphs if and only if $\mathrm{EP}^+(q-1)$ does.*

Proof. Both graphs below remain strong C_4-free, so $S = 2e - 3T$ and the deletion loss of a vertex set is determined by the edges and triangles that meet it.

First assume $\mathrm{EP}^+(q-1)$, and let (u, v) be an oriented edge of maximum induced-path order q, with terminal set E. Delete $W = N[u] \setminus \{v\}$, and let H be the component of v in $G - W$. An induced path of G that starts with uv has no further vertex in $N[u]$, because the path is induced, so the remainder is an induced path of H starting at v. Conversely, every induced path of H starting at v extends by u to an induced path of G, since no vertex of H other than v is adjacent to u. Thus $p_v(H) = q - 1$, and the terminal set in H is still E.

Each $x \in E$ has at most one neighbour in W: any such neighbour is adjacent to u, and two of them would be two common neighbours of u and x. Every neighbour of x outside H lies in W. Let m_W be the number of edges between E and W, so $m_W \le |E|$, and let T_{extra} be the number of triangles that meet E and are not contained in H. Every such triangle uses one of those edges. Comparing edges and triangles,

$$D_G(E) = D_H(E) + 2m_W - 3T_{\mathrm{extra}} \le D_H(E) + 2|E|.$$

The rooted inequality in H is $D_H(E) \le (q-2)|E|$, because $\mathrm{EP}^+(q-1)$ contributes the factor $(q-1)-1$. Therefore $D_G(E) \le q|E|$.

Conversely, assume $\mathrm{EP}(q)$, and let (H, v) be a connected strong C_4-free graph with $p_v(H) = q-1$ and terminal set E. Form G by adding a vertex u adjacent to v and, for each $x \in E$, a new vertex a_x adjacent exactly to u and to x. Every terminal x fails to be adjacent to v: an induced path of order $q-1 \ge 3$ from v to x has no chord vx. The vertex a_x has degree two. Any cycle through a_x uses both edges ua_x and $a_x x$, and then continues from x to the other neighbour of u on the cycle. That other neighbour is v or some a_y. In the first subcase the x–v path in H has length at least two, so the cycle has length at least five. In the second the cycle has the form $u, a_x, x, \dots, y, a_y$ and again has length at least five. A cycle through u that avoids every a_x would need two neighbours of u in H, but the only neighbour of u in H is v. Thus G is strong C_4-free. No new triangle appears: the two neighbours of a_x are nonadjacent, and the neighbours of u are pairwise nonadjacent.

An induced path that starts with uv contains no a_x. Otherwise the edge ua_x would be a chord, since a_x is not the successor of u on that path. The induced uv-paths are therefore the paths u followed by an induced path of H starting at v. Their maximum order is q, and their terminal set is E. Exactly $|E|$ new edges meet E, namely the edges $a_x x$, and no new triangle meets E. Hence $D_G(E) = D_H(E) + 2|E|$. The oriented inequality gives $D_G(E) \le q|E|$, so $D_H(E) \le (q-2)|E|$, which is $\mathrm{EP}^+(q-1)$. □

Example S3.4. $\mathrm{EP}(9)$ fails. Let G have vertices $0, \dots, 14$ and adjacency lists

$$\begin{array}{llll} 0\colon 4,8,11 & 1\colon 5,8,10 & 2\colon 6,8,9 & 3\colon 7,10 \\ 4\colon 0,9,10 & 5\colon 1,9,11 & 6\colon 2,10,11 & 7\colon 3,11 \\ 8\colon 0,1,2,12 & 9\colon 2,4,5 & 10\colon 1,3,4,6,13 & 11\colon 0,5,6,7,14 \\ 12\colon 8,13,14 & 13\colon 10,12 & 14\colon 11,12. & \end{array}$$

The graph is triangle-free and strong C_4-free, with $n = 15$, $e = 23$ and $S = 46$. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by

$$12,8,2,9,4,10,3,7,11 \qquad \text{and} \qquad 12,8,2,9,5,11,7,3,10.$$

Thus $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. No induced path that starts with $12, 8$ has order greater than 9. The neighbours of 8 other than 12 are $0, 1, 2$. A continuation is legal only when the new vertex meets the current end and meets no earlier vertex of the path. Under that rule every maximal induced path is one of the following, and the only two of order 9 are the paths already displayed.

maximal induced path starting 12, 8	order
12, 8, 0, 4, 9, 5	6
12, 8, 0, 4, 10, 3, 7	7
12, 8, 0, 4, 10, 6	6
12, 8, 0, 11, 5, 9	6
12, 8, 0, 11, 6, 10, 3	7
12, 8, 0, 11, 7, 3, 10	7
12, 8, 1, 5, 9, 4	6
12, 8, 1, 5, 11, 6	6
12, 8, 1, 5, 11, 7, 3	7
12, 8, 1, 10, 3, 7, 11	7
12, 8, 1, 10, 4, 9	6
12, 8, 1, 10, 6, 11, 7	7
12, 8, 2, 6, 10, 3, 7	7
12, 8, 2, 6, 10, 4	6
12, 8, 2, 6, 11, 5	6
12, 8, 2, 6, 11, 7, 3	7
12, 8, 2, 9, 4, 10, 3, 7, 11	9
12, 8, 2, 9, 5, 11, 7, 3, 10	9

The first of the two paths of order 9 ends at 11. The neighbours of 11 are 0, 5, 6, 7, 14: the vertex 7 is the predecessor, 0 meets 8, 5 meets 9, 6 meets both 2 and 10, and 14 meets 12. The second path ends at 10. The neighbours of 10 are 1, 3, 4, 6, 13: the vertex 3 is the predecessor, 1 meets 8, 4 meets 9, 6 meets both 2 and 11, and 13 meets 12. Neither path extends. The same computation gives a failure of $\mathrm{EP}^{+}(8)$ on the subgraph induced by $0, \dots, 11$, rooted at 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. The inequalities EP(8) and $\mathrm{EP}^{+}(6)$ are not settled here.

Example S3.5. Let $k = p - r$. Call an induced subgraph H of G an *eligible core* if it is proper, connected and dominating, and $p(H) \le p(G) - 2$. The budget CB asks for an eligible core with

$$S(G) - S(H) \le k\big(n - |H|\big) + |H|.$$

Together with REAL-133 for H and an induction on the order, the budget would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$ and $k = 3$, hence $\ell = 3$. An eligible core has $p(H) \le 3$. A connected triangle-free strong C_4-free P_4-free graph is a star, by Lemma 2.7 and the absence of triangles. In the Petersen graph a star with $k \le 3$ leaves has centre of degree 3. The vertices it dominates are the centre, its three neighbours, and at most two further neighbours of each leaf, at most $4 + 2k$ vertices in all. This is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$, and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$.
The unrestricted budget therefore fails. The example has $\ell = 3$.

Example S3.6. The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, and $Z = 28$ nonadjacent pairs with no common neighbour) satisfy the counting identities used above, including $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$, together with the lower bounds $p \ge 2r - 1$ and $p \ge \min_v \lambda(v) + 2$, if p is replaced by the value 5. But $5 < r + \ell = 6$. This system of scalar constraints does not imply REAL-133. The Heawood graph itself has $p = 7$.

Example S3.7. Let q be a prime power, and let F be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2 + q + 1)$ and $S(F) = (q+1)f$, and choose a root of F. Let $t = f - 1$ and $b \ge 1$. Take a 6-cycle $c_0 \cdots c_5$, attach a pendant path with t edges at each cycle vertex, and attach b copies of F at c_0 by identifying their roots with c_0. The result is strong C_4-free, because each new block meets the rest in a cut vertex, and

$$n = 6f + b(f-1), \qquad S = 12f + b(q+1)f, \qquad r = t + 3, \qquad p = 2t + 5 = 2r - 1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f - 1 \le t$ of c_0, so each cycle vertex has eccentricity $t + 3$, while a vertex x of a copy has distance $d_F(x, c_0) + t + 3$ from the tip of the arm at c_3. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most t vertices. Mass is additive at cut vertices by Lemma 2.3. For fixed q, as $b \to \infty$ one has $\ell \to (q+1)f/(f-1)$. Letting $q \to \infty$, with $r = f + 2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy RAD(r) for their own radius, and they show that ℓ is unbounded on the class $p = 2r - 1$. Any theorem RAD(r) must use a bound that depends on r.

4 | Candidate pairs above density four

The two lemmas below are the density restrictions for induced-path order 7 and for bipartite graphs of radius 2. The pairs $(r, p) = (2, 6)$ and $(3, 6)$ are settled by Corollary 6.3.

Lemma S4.1. *Let G be connected, bipartite and strong C_4-free, with $r(G) = 2$. Then G is a tree, $\ell(G) = 2-2/n$ and $p(G) \le 5$.*

Proof. Let s be a centre, and let A be the colour class of s and B the other class. Every vertex of B lies at odd distance from s, hence at distance 1, so $B \subseteq N(s)$. Every vertex $x \in A \setminus \{s\}$ lies at distance 2 from s, so x has a neighbour in B. Two such neighbours would form a 4-cycle with s. Thus x has exactly one neighbour. There is no edge inside A or inside B, and the number of edges is $|B| + |A| - 1 = n - 1$. A connected graph with $n - 1$ edges is a tree. There are no triangles, so $S = 2e = 2n - 2$ and $\ell = 2 - 2/n$. Every vertex of $A \setminus \{s\}$ is a leaf. An induced path contains at most two leaves, and the unique path between them has the form leaf, neighbour in B, s, neighbour in B, leaf. Hence $p(G) \le 5$. □

Lemma S4.2. *Let G be a connected triangle-free strong C_4-free P_8-free graph. Then $e(G) \le 2n - 2$. In particular, for $n \ge 2$ one has $\ell(G) = 2e(G)/n < 4$.*

Proof. The graph has no triangle and no 4-cycle subgraph, so every cycle has length at least 5. The bound $e \le 2n-2$ holds for $n = 1$. Let $n \ge 2$, and proceed by induction on the order. If G is a tree, then $e = n-1 \le 2n-2$. Otherwise the girth is at least 5. Theorem 1 of Chudnovsky and Stacho [3] then asserts that G is the Petersen graph, the Heawood graph, the graph obtained by contracting one edge of the Heawood graph, or else G has a vertex of degree at most 2. These three exceptions have pairs (n, e) equal to $(10, 15)$, $(14, 21)$ and $(13, 20)$. Each satisfies $e \le 2n - 2$, since $15 \le 18$, $21 \le 26$ and $20 \le 24$.
In every remaining case some vertex v has degree at most 2. Every component H of $G - v$ is connected, triangle-free, strong C_4-free and P_8-free. The inductive hypothesis gives $e(H) \le 2|H| - 2$. If $G - v$ has $c \ge 1$ components, then

$$e(G) = e(G - v) + d(v) \le 2(n - 1) - 2c + 2 \le 2n - 2,$$

because $d(v) \le 2$ and $c \ge 1$. Since G is triangle-free, $S = 2e \le 4n - 4$, so $\ell \le 4 - 4/n < 4$. □

If $p = 6$, Corollary 6.3 gives $S \le 3n-1$ and $\ell < 3$. The pairs $(r, p) = (2, 6)$ and $(3, 6)$ are therefore excluded from the range $\ell \ge 3$. The same corollary shows that 3 is a sharp supremum and is not attained: the graph F_2 of Theorem 5.6, with at least three branches of length 3, has radius 2 and $\ell = 3-5/n$, and the graph θ_k of Proposition 6.2 has radius 3 and $\ell = 3-6/n$. Lemma S4.2 gives $\ell < 4$ only under the triangle-free hypothesis. The lemma applies to every connected triangle-free strong C_4-free graph of induced-path order 7, since such a graph is P_8-free. If $p = 7$ and $\ell \ge 4$, then G contains a triangle. In the range $5 < \ell \le 6$ the pairs $(2, 7)$ and $(3, 7)$ therefore occur only for graphs that contain a triangle. Example S4.3 is a triangle-free graph with $(r, p) = (2, 7)$ and $\ell = 17/6 < 4$. Lemma S4.1 is the bipartite restriction: a connected bipartite strong C_4-free graph of radius 2 is a tree, so its induced-path order is at most 5.

Example S4.3. Let P be the Petersen graph, and let u and v be the ends of an induced P_5 in P. Such a path exists by the proof of Theorem 5.6, and u and v are nonadjacent. Form G by adding one leaf at u and one leaf at v. Then

$$(r(G), p(G), |V(G)|, S(G), \ell(G)) = (2, 7, 12, 34, 17/6).$$

The graph P has 15 edges. The two new edges create no cycle, so G is triangle-free and strong C_4-free, $n = 12$, $e = 17$ and $S = 2e = 34$. Thus $\ell = 17/6$. The graph P is P_6-free, because it is the extremal graph of order 10 in Theorem 3.6. A leaf has degree one, so it is an end of every induced path that contains it. Deleting the leaves that lie on an induced path of G leaves an induced path of P, of order at most 5. The path formed by the two leaves and the chosen P_5 is induced, because each leaf is adjacent only to its endpoint. Hence $p(G) = 7$. The radius is 2. The proof of Theorem 3.6 records that the diameter of P is 2, so u and v have a common neighbour w. Every vertex of P is at distance at most 2 from w, and each new leaf is at distance 2 from w. Thus the eccentricity of w is 2.

Suppose $4 < \ell \le 5$ and G violates REAL-133. Then $\lceil \ell \rceil = 5$, so Corollary 6.1 with $K = 4$ gives either $r \le 4$ and $2r - 1 \le p \le r + 4$, or $r = 5$ and $p = 9$. Theorem 5.8 forces $\ell \le 3$ whenever $p \le 6$, so $p \ge 7$. Corollary 4.5 forces $\ell \le 2$ when $(r, p) = (4, 7)$. The pairs that remain are $(3, 7)$, $(4, 8)$ and $(5, 9)$. The candidate $(3, 7)$ must contain a triangle: Lemma S4.2 gives $\ell < 4$ for every connected triangle-free strong C_4-free graph of induced-path order 7.
For $5 < \ell \le 6$, Corollary 6.1 gives $2r - 1 \le p \le r + 5$ and $r \le 6$. Lemma 3.1 excludes radius 1, Theorem 3.6 excludes $p \le 5$, and Corollary 4.5 excludes $(4, 7)$. Corollary 6.3 excludes $(2, 6)$ and $(3, 6)$ from $\ell \ge 3$. Lemma S4.2 uses the triangle-free hypothesis: if $p = 7$ and $\ell \ge 4$, then G contains a triangle, so $(2, 7)$ and $(3, 7)$ occur in this range only for graphs that contain a triangle.

Statement	Status
ROOT(3) and ROOT(4); the terminal strengthening of Proposition S1.3	proved, Lemma 5.3 and Theorem S1.2
ROOT(q) for every q, in the class $\mathcal{B}$	proved, Theorem S2.2
a minimum counterexample to GEN_4 is 2-connected	proved, Proposition S1.4; GEN_4 is open
GEN_K for $K \geq 4$, unrestricted GEN_6, RAD(r) for $r \geq 5$, and ROOT(q) for $q \geq 5$ outside $\mathcal{B}$	open
pairs (r, p) still necessary for $4 < \ell \leq 5$	$(3,7)$, $(4,8)$, $(5,9)$; the candidate $(3,7)$ contains a triangle, by Lemma S4.2
pairs (r, p) still necessary for $5 < \ell \leq 6$	$(2,7)$, $(3,7)$, $(3,8)$, $(4,8)$, $(4,9)$, $(5,9)$, $(5,10)$, $(6,11)$
$(2,6)$ and $(3,6)$	$\ell < 3$, and 3 is a sharp unattained supremum: F_2 with $t \geq 2$ gives $3 - 5/n$ at radius 2, and θ_k gives $3 - 6/n$ at radius 3, Corollary 6.3
$(2,7)$ and $(3,7)$ in the range $5 < \ell \leq 6$	each such graph contains a triangle; the bound $\ell < 4$ of Lemma S4.2 uses the triangle-free hypothesis
$(r, p) = (2,7)$ in a triangle-free graph	possible with $\ell < 4$; Example S4.3

TABLE 1 | Rooted bounds and the candidate pairs above density four. The rows for $4 < \ell \leq 5$ and for $5 < \ell \leq 6$ record necessary pairs only. For $4 < \ell \leq 5$ the candidate $(3,7)$ contains a triangle, by Lemma S4.2.

Conjecture S4.4. ROOT(q) *holds for every* $q \geq 3$. *For every* $r \geq 5$, *every connected strong* C_4*-free graph of radius* r *and path order* $2r-1$ *satisfies* $S \leq (r-1)n - 2r(r-3)$, *with equality only for* C_{2r}.

The cases $q = 3$ and $q = 4$ are Lemma 5.3 and Theorem S1.2, and every q holds in $\mathcal{B}$ by Theorem S2.2. The radius bound for $r \geq 5$ is open.

Conjecture S4.5. *If* F *is connected and strong* C_4*-free,* $p_s(F) = q$ *and the eccentricity of* s *is* $q-1$, *then*

$$S(F) \leq (q-1)(n-1) - (q-3)D - m - (q-2)(q-3).$$

For $q = 4$ the subtracted term is 2, and Proposition S1.3 is this case. For $q \geq 5$ the conjecture is open. The terminal bound in $\mathcal{B}$ is the first part of Theorem S2.2.

Conjecture S4.5, for a fixed q, implies ROOT(q). Let F be connected and strong C_4-free with $p_s(F) \leq q$. Form G by attaching a path of $q-1$ new edges at s. Then G is terminal of index q: the new path is an induced path of order q starting at s, and every induced path of F that starts at s has order at most q. The order, the root degree and the mass increase by $q-1$, 1 and $2(q-1)$ respectively, and m is unchanged. Write $N = n + q - 1$ and $D_G = D + 1$. The terminal inequality in G reads

$$S(F) + 2(q-1) \leq (q-1)(N-1) - (q-3)D_G - m - (q-2)(q-3).$$

The identity $N - 1 = n + q - 2$ gives $(q-1)(N-1) = (q-1)(n-1) + (q-1)^2$, and $(q-3)D_G = (q-3)D + (q-3)$. The terms that do not involve n, D or m are

$$(q-1)^2 - (q-3) - (q-2)(q-3) - 2(q-1) = (q-3)\big((q-1) - (q-1)\big) = 0.$$

What remains is $S(F) \leq (q-1)(n-1) - (q-3)D - m$, which is ROOT($q$). For $q = 4$ this route uses Proposition S1.3. The induction in Theorem S1.2 does not. For $q \geq 5$ both the hypothesis and ROOT(q) are open.

Conjecture S4.6. *Every connected strong* C_4*-free graph* G *that is not a cycle and has* $\ell(G) > 5$ *has an eligible core* H, *in the sense of Example S3.5, with*

$$S(G) - S(H) \leq (p-r)\big(n - |H|\big) + |H|.$$

Example S3.5 is the failure of the unrestricted budget at the Petersen graph, where $\ell = 3$. Example S3.6 meets the listed scalar constraints when the path order is replaced by 5, while $5 < r + \ell$, and the Heawood graph itself has $p = 7$.

5 | One triangle and the edge deficit

Theorem S5.1 is the hand proof for a graph with at most one triangle. Lemma S5.2 classifies the ways two triangles can meet, and Lemma S5.3 is the companion finite search. Lemmas S5.5, S5.6 and S5.7 bound the number of edges through the deficit $\Delta(G) = 2n - 3 - e(G)$.

Theorem S5.1. *Let G be a connected strong C_4-free P_7-free graph with at most one triangle. Then $S(G) \le 3n$. In particular the same bound holds for a graph with exactly one triangle.*

Proof. Suppose G is a counterexample of minimum order in the stated class, so $S(G) \ge 3n+1$. Theorem 5.5 forces G to contain a triangle, hence exactly one. Let v be a cut vertex. For each component C of $G-v$ let ρ_C be the largest order of an induced path in $G[C\cup\{v\}]$ with end v. Two such paths from different components form an induced path on $\rho_C+\rho_{C'}-1 \le 6$ vertices, so some component C_1 has $\rho_{C_1} \le 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Both graphs are connected, smaller than G, strong C_4-free and P_7-free, and each has at most one triangle. Lemma 5.3 gives $S(G_1) \le 2(n_1-1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3,

$$S(G) \le 2(n_1-1) + 3n_2 = 3n - (n_1-1) < 3n.$$

Thus G is 2-connected.
Let v be a vertex of minimum degree among the three vertices of the triangle, write $d = d(v)$, and let u and w be the other two. The graph $H = G - v$ is connected and triangle-free. Apply Lemma 5.4 to H, and write Q for its core, $h = |V(Q)|$ and $\eta = \eta(Q)$. Set

$$a = \big|N_G(v) \cap V(Q)\big|, \qquad c = \big|V(H) \setminus \big(V(Q) \cup N_G(v)\big)\big|,$$

and $t = |V(H)| - h$. The vertices of H outside Q are the $d-a$ neighbours of v that miss Q, together with the c vertices counted above, so $t = d - a + c$. Lemma 5.4 gives $e(H) = e(Q) + t$. Since G has one triangle, $S(G) = 2e(G) - 3$ and $e(G) = e(H) + d$, and therefore

$$S(G) - 3n = 2d - 6 - \eta - t = d + a - 6 - \eta - c. \tag{5.1}$$

The core satisfies $a \le \eta + 3$ and, for every edge xy of Q,

$$a + \min\{d_Q(x), d_Q(y)\} \le \eta + 5. \tag{5.2}$$

These bounds are read from the list in Lemma 5.2, using the descriptions that precede that lemma. If Q is the Petersen graph, then $\eta = 0$. Every two nonadjacent vertices of the Petersen graph have a common neighbour, so two such vertices cannot both be adjacent to v. Thus $N(v) \cap V(Q)$ is a clique, necessarily of size at most two, and every degree in Q is three. Both bounds follow. For each remaining fixed graph of order at least three one has $h \le 11$ and minimum degree at least two. The open neighbourhoods in Q of any two vertices of $N(v) \cap V(Q)$ are disjoint: a common neighbour would form a 4-cycle with v. Hence $a \le \lfloor h/2 \rfloor \le 5$. Every degree in these graphs is at most four. The only rows with $\eta < 4$ are F_4 and F_{12}, both with $\eta = 3$. The path-and-spoke description of F_4 has maximum degree three. In F_{12} the only vertices of degree four are v_1 and v_5 on the 7-cycle. Each is adjacent to two cycle vertices and to two of $\{x_1, x_2, x_5\}$, and v_1 is not adjacent to v_5, so every edge of F_{12} has an end of degree at most three. This gives (5.2) and the bound on a. In each parameterised family every non-hub vertex is adjacent to a hub, and each hub has at most one neighbour in common with v, so $a \le 4$ even after both hubs are counted. Every edge has an end of degree two, because the edges are the internal edges of the paths between the hubs described above, and η is 5 or 6. The graphs K_1 and K_2 satisfy both bounds directly: K_1 has $\eta = 3$ and no edge, and K_2 has $\eta = 4$.
If u or w lies outside Q, say u, the tree attachments give u at most one neighbour in Q. Apart from w, no neighbour of u in H is adjacent to v, because the triangle is unique. Thus $d \le d(u) \le c + 3$. Substituting in (5.1) gives $S(G) - 3n \le a - \eta - 3$, and $a \le \eta + 3$ yields $S(G) - 3n \le 0$. If instead both u and w lie in Q, then uw is an edge of Q, and every neighbour of either vertex that lies in $H - V(Q)$ lies in the set counted by c. Since v has minimum degree among the three triangle vertices,

$$d \le 1 + c + \min\{d_Q(u), d_Q(w)\}.$$

Substituting in (5.1) and applying (5.2) to the edge uw again yields $S(G) \le 3n$. □

Lemma S5.2. *Let G be strong C_4-free, and let T_1 and T_2 be triangles of G.*

1. *Suppose T_1 and T_2 share a vertex u, and write $T_1 = \{u, a, b\}$ and $T_2 = \{u, c, d\}$. If ac is an edge, then $a\ c\ d\ u$ is a 4-cycle. The same holds for each of the other three edges with one end in $\{a, b\}$ and one end in $\{c, d\}$.*
2. *Suppose T_1 and T_2 are disjoint, and let xy and $x'y'$ be distinct edges with $x, x' \in T_1$ and $y, y' \in T_2$. If the two edges share an endpoint in T_1, say $x = x'$ and $y \ne y'$, and if t is the third vertex of T_2, then $x\ y\ t\ y'$ is a 4-cycle. If they share an endpoint in T_2, the same construction with the roles of T_1 and T_2 exchanged is a 4-cycle. If they share no endpoint, then $x \ne x'$ and $y \ne y'$, both pairs are edges of their triangles, and $x\ y\ y'\ x'$ is a 4-cycle. In particular there is at most one edge with one end in T_1 and one end in T_2.*

3. *Suppose T_1 and T_2 are disjoint and lie in the same component. Let d be the distance between their vertex sets, and let $P : u = p_0, p_1, \dots, p_d = v$ be a path of length d with $u \in T_1$ and $v \in T_2$. The internal vertices of P lie outside $T_1 \cup T_2$, and the subgraph induced by $V(T_1) \cup V(T_2) \cup V(P)$ consists of the edges of T_1, the edges of T_2 and the edges of P.*
4. *In the notation of (3), if $x \in T_1 \setminus \{u\}$ and $z \in T_2 \setminus \{v\}$, then $x, u, p_1, \dots, p_{d-1}, v, z$ is an induced path on $d + 3$ vertices.*

Proof. (1) The vertices a, c, d and u are distinct: T_1 and T_2 share only u, so $a \notin T_2$. The edges ac, cd, du and ua are present, the last three because cd and du lie in T_2 and ua lies in T_1. This is the 4-cycle $a\,c\,d\,u$. Each of the other three cross edges is the same configuration after renaming.
(2) First suppose $x = x'$ and $y \neq y'$. Then y and y' are distinct vertices of the triangle T_2, so both are adjacent to the third vertex t of T_2. The vertices x, y, t, y' are distinct because $x \notin T_2$. The edges xy, yt, ty' and $y'x$ are present, and $x\,y\,t\,y'$ is a 4-cycle. The case of a shared endpoint in T_2 is the same argument with the triangles exchanged.
Now suppose the four endpoints are distinct. Any two vertices of a triangle are adjacent, so xx' and yy' are edges. The vertices x, y, y', x' are distinct because the triangles are disjoint. The edges xy, yy', $y'x'$ and $x'x$ are present, and $x\,y\,y'\,x'$ is a 4-cycle. A second edge between T_1 and T_2 falls under one of these two cases, so a strong C_4-free graph has at most one such edge.
(3) A shortest path between the two sets meets each set only at an endpoint, so every internal vertex lies outside $T_1 \cup T_2$. A chord of P would give a shorter u–v path.
Let $x \in T_1 \setminus \{u\}$, and let y be the third vertex of T_1, so $\{u, x, y\}$ is a triangle. Suppose x is adjacent to p_i for some $i \in \{1, \dots, d\}$. If $i = 1$, the vertices y, u, p_1, x are distinct and the edges yu, up_1, p_1x and xy are present, so $y\,u\,p_1\,x$ is a 4-cycle. If $i \geq 2$, the walk $x, p_i, p_{i+1}, \dots, p_d$ has length $d - i + 1 \leq d - 1$ and joins T_1 to T_2, contradicting the definition of d. Thus x has no neighbour on P except u. The same argument, using the third vertex of T_2, shows that no vertex of $T_2 \setminus \{v\}$ has a neighbour on P except v. In particular, adjacency to p_{d-1} produces a 4-cycle, and adjacency to an earlier vertex of P shortens the distance.
It remains to rule out an edge with one end in T_1 and one end in T_2, other than the edge uv in the case $d = 1$. If $d \geq 2$, any such edge would be a path of length 1 between the two sets. If $d = 1$, the path P is the edge uv, and a second edge is a 4-cycle by (2).
No other pair of vertices in $V(T_1) \cup V(T_2) \cup V(P)$ remains. The induced subgraph is therefore the two triangles joined by P.
(4) The named walk has $d + 3$ vertices and uses edges of T_1, of P and of T_2. A chord incident with x would be an edge from x to a vertex of the walk other than u. Part (3) forbids every such edge, including an edge from x to z. A chord incident with z is forbidden in the same way. A chord that avoids x and z is a chord of P or an edge forbidden by (3). The walk is an induced path. □

Lemma S5.3. *Every connected strong C_4-free P_7-free graph with at least two triangles has a vertex v with $L(v) \leq 3$.*

Proof. Let G be connected, strong C_4-free and P_7-free, and suppose G has at least two triangles. Two triangles do not share an edge. Suppose they share the edge uv, and let x and y be the third vertices. The vertices x, u, y, v are distinct, and the edges xu, uy, yv and vx are present, because each of x and y is adjacent to both ends of uv. The cycle $x\,u\,y\,v$ is a 4-cycle. Thus two triangles share at most one vertex.
If two triangles share a vertex, Lemma S5.2(1) forbids an edge between their unshared vertices. Their union is an induced subgraph on five vertices. If the triangles are disjoint, let d be the distance between their vertex sets. Lemma S5.2(3) identifies the induced subgraph formed by the two triangles and a shortest connecting path, and Lemma S5.2(4) supplies an induced path on $d + 3$ vertices. Since $p(G) \leq 6$, one has $d \leq 3$.
Four induced subgraphs therefore exhaust the configurations. Two triangles may share a vertex, or they may be joined by a path of length 1, 2 or 3. These graphs have orders 5, 6, 7 and 8 respectively. From each of them, grow a labelled induced subgraph by the rule of Lemma 5.7, with one change: at every node select a single vertex whose current deletion loss is strictly less than 4, and admit only those neighbourhoods that meet the corresponding one-vertex set. Section A records the four searches. They visit 36, 31, 46 and 5 nodes, and the largest orders they reach are 17, 15, 11 and 10. Every branch closes. An induced copy of any of the four graphs inside a graph with $L \geq 4$ at every vertex would extend at every node. Therefore no graph with at least two triangles has $L \geq 4$ at every vertex. □

Remark S5.4. Lemma S5.3 yields a second proof of the inequality $S \leq 3n$. Proceed by induction on the order. Theorem 5.5 and Theorem S5.1 give the inequality when the graph has at most one triangle. If there are at least two triangles, Lemma S5.3 supplies a vertex v with $L(v) \leq 3$. Each component of $G - v$ is smaller and lies in the class, so the inductive bound sums to $S(G - v) \leq 3(n - 1)$. Hence $S(G) \leq 3(n - 1) + 3 = 3n$. The equality analysis remains that of Theorem 5.8.

Lemma S5.5. *Let G be a graph and let K be a clique cutset of order t, with $G-K$ having $k \geq 2$ components. For each component C let G_C be the subgraph induced by $C \cup K$, and write $\Delta(H) = 2|V(H)| - 3 - e(H)$. Then*

$$\Delta(G) = \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].$$

The bracket equals 1 if $t = 1$ and equals 0 if $t = 2$ or $t = 3$.

Proof. The edges inside K are counted k times among the graphs G_C, and $|V(G)| = \sum_C |V(G_C)| - (k-1)t$, so

$$e(G) = \sum_C e(G_C) - (k-1)\binom{t}{2}.$$

Therefore

$$\begin{aligned}\Delta(G) &= 2\sum_C |V(G_C)| - 2(k-1)t - 3 - \sum_C e(G_C) + (k-1)\binom{t}{2} \\ &= \sum_C \big(\Delta(G_C) + 3\big) - 3 - 2(k-1)t + (k-1)\binom{t}{2} \\ &= \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].\end{aligned}$$

The bracket is $0+3-2=1$ for $t=1$, $1+3-4=0$ for $t=2$, and $3+3-6=0$ for $t=3$. □

Lemma S5.6. *Let G be a connected strong C_4-free P_7-free graph on $n \geq 2$ vertices, and write $\Delta(G) = 2n - 3 - e(G)$. Then $\Delta(G) \geq 0$. If $n \geq 4$, then $\Delta(G) \geq 1$, so $e(G) \leq 2n-4$ and $S(G) \leq 4n - 8 - 3T(G)$. In particular $\ell(G) < 4$.*

Proof. A copy of K_4 contains a 4-cycle, so $\omega(G) \leq 3$. A diamond contains a 4-cycle, so every graph in the class is $(P_7, C_4, \text{diamond})$-free in the sense of Chen, Wu and Xu [2]. The proof is by induction on n. For $n = 2$ the graph is K_2 and $\Delta = 0$. For $n = 3$ the maximum of e is 3, attained by K_3, and $\Delta \geq 0$.

Let $n = 4$. A graph on four vertices with five edges is K_4 minus an edge. Labelling the missing edge cd, the cycle $a\,c\,b\,d$ uses four edges of the graph. Thus $e \leq 4$ and $\Delta = 5 - e \geq 1$.

Now let $n \geq 5$, and assume the two claims for every smaller graph in the class. Every block G_C arising from a clique cutset is itself connected, strong C_4-free and P_7-free, of order strictly less than n.

Suppose first that v is a cut vertex, and that $G - v$ has $k \geq 2$ components. Lemma S5.5 with $t = 1$ gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. Each side has order at least 2, so $\Delta(G_i) \geq 0$ by induction, and $\Delta(G) \geq k - 1 \geq 1$.

Suppose next that a triangle K is a clique cutset and $G - K$ has $k \geq 2$ components. Each side G_i contains K and at least one further vertex, so its order is at least 4. Lemma S5.5 gives $\Delta(G) = \sum_i \Delta(G_i)$, and induction supplies $\Delta(G_i) \geq 1$. Hence $\Delta(G) \geq k \geq 2$.

Suppose that $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. Lemma S5.5 gives $\Delta(G) = \sum_i \Delta(G_i)$. Induction gives $\Delta(G_i) \geq 0$ for every side, and each side has order at least 3. If every side had deficit 0, then each side would have order 3 and exactly three edges, so each side would be a triangle. Two triangles that share the edge uv span a 4-cycle on their two opposite vertices, which contradicts the hypothesis that G is strong C_4-free. Some side therefore has positive deficit, and $\Delta(G) \geq 1$.

Suppose finally that G has no clique cutset. Theorem 1.1 of [2] states that a connected $(P_7, C_4, \text{diamond})$-free graph with no clique cutset is the Petersen graph, or the graph F defined below, or else has minimum degree at most $\max\{2, \omega - 1\}$. Here $\omega \leq 3$, so the last quantity is at most 2. The Petersen graph has $n = 10$ and $e = 15$, hence $\Delta = 2$. The graph F is obtained from a 7-cycle $v_1 \cdots v_7$ by adding an independent set $\{y_1, y_2, y_3\}$ with

$$N(y_1) = \{v_1, v_4, v_5\}, \qquad N(y_2) = \{v_2, v_5, v_6\}, \qquad N(y_3) = \{v_3, v_6, v_7\}.$$

The three triangles of F are $y_1v_4v_5$, $y_2v_5v_6$ and $y_3v_6v_7$, so $T(F) = 3$. Since $n = 10$ and $e = 16$, one has $\Delta(F) = 1$. The path $v_1, v_7, y_3, v_3, v_4, v_5, y_2$ is an induced P_7, so F lies outside the class. The deficit bound does not need F as an exception: $\Delta(F) = 1$ meets the lower bound for $n \geq 4$. Every remaining graph in the class with no clique cutset is the Petersen graph or has a vertex of degree at most 2. If some vertex v has degree at most 2, then v is not a cut vertex, so $G - v$ is connected and lies in the class. Deleting v changes the deficit by

$$\Delta(G) - \Delta(G - v) = 2 - d(v) \geq 0.$$

For $n \geq 5$ the order of $G - v$ is at least 4, so induction gives $\Delta(G - v) \geq 1$ and therefore $\Delta(G) \geq 1$.
Lemma 2.1(3) gives $S = 2e - 3T$. For $n \geq 4$ one has $e \leq 2n - 4$, so $S \leq 4n - 8 - 3T \leq 4n - 8$ and $\ell \leq 4 - 8/n < 4$. For $2 \leq n \leq 3$ one has $e \leq 2n - 3$ and $S \leq 4n - 6$, so again $\ell < 4$. □

Lemma S5.7. *Let G be a connected strong C_4-free P_7-free graph.*

1. *If $n \geq 5$ and $T(G) \leq 1$, then $\Delta(G) \geq 2$, so $e(G) \leq 2n - 5$. If $T(G) = 1$, then $S(G) \leq 4n - 13$.*
2. *If $n \geq 4$ and $T(G) \geq 2$, then $S(G) \leq 4n - 14$.*
3. *If $n \geq 5$ and $T(G) \geq 1$, then $S(G) \leq 4n - 13$.*
4. *If $n \geq 6$ and $T(G) \leq 2$, then $\Delta(G) \geq 2$.*

Proof. If $T \geq 2$ and $n \geq 4$, Lemma S5.6 gives $e \leq 2n - 4$, hence $S = 2e - 3T \leq 2(2n - 4) - 6 = 4n - 14$. This is (2). Once (1) is known, $T = 1$ gives $S = 2e - 3 \leq 4n - 13$, and (2) is at least as strong, so (3) follows. It remains to prove $\Delta \geq 2$ when $n \geq 5$ and $T \leq 1$.
The proof is by induction on n. Let $n = 5$. A hole is a chordless cycle of length at least four. In a strong C_4-free graph every hole has length at least 5. The only possible hole on five vertices is a 5-cycle. A chord of C_5 produces a 4-cycle, and there is no room for a further vertex, so $G \cong C_5$. Then $e = 5$, $T = 0$ and $\Delta = 2$. If G has no hole, then G is chordal. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and its neighbourhood is a clique. That clique has order at most two: a simplicial vertex of degree three or more, together with three neighbours, contains a K_4 and therefore a 4-cycle. Deleting a simplicial vertex of degree one removes one edge and no triangle. Deleting a simplicial vertex of degree two removes two edges and exactly one triangle. Repeating until a single vertex remains gives $e = n - 1 + T$. Here $T \leq 1$, so $e \leq 5$ and $\Delta = 7 - e \geq 2$.
Now let $n \geq 6$, with $T(G) \leq 1$, and assume the deficit claim for every smaller graph in the class that has at most one triangle. By Lemma S5.6 every graph in the class of order at least 2 has nonnegative deficit, and every graph of order at least 4 has deficit at least 1.
Suppose v is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma S5.5 gives $\Delta(G) = \sum_i \Delta(G_i) + (k - 1)$. If this quantity were at most 1, then $k = 2$ and both sides would have deficit 0. A side of deficit 0 has order at most 3, so $n \leq 3 + 3 - 1 = 5$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose a triangle is a clique cutset, with $k \geq 2$ sides. Each side has order at least 4, so each has deficit at least 1 by Lemma S5.6, and Lemma S5.5 gives $\Delta(G) \geq 2$.
Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components, and suppose toward a contradiction that $\Delta(G) \leq 1$. Lemma S5.5 gives $\Delta(G) = \sum_i \Delta(G_i)$. A side of order at least 4 has deficit at least 1. A side of deficit 0 therefore has order 3 and three edges, so it is a triangle. Two such sides would give two triangles and a 4-cycle, which is impossible under $T(G) \leq 1$ and the strong C_4-free hypothesis. If every side had deficit at least 1, the sum would be at least $k \geq 2$. Hence exactly one side is a triangle, of deficit 0, and exactly one other side is present: a third side would contribute deficit at least 1 and would push the sum to at least 2. The hypothesis $T(G) \leq 1$ forces the other side G_2 to be triangle-free. Its order is $n - 1 \geq 5$. The inductive hypothesis gives $\Delta(G_2) \geq 2$, so $\Delta(G) \geq 2$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose finally that G has no clique cutset. The Petersen graph has $T = 0$ and $\Delta = 2$. The graph F of Lemma S5.6 has $T(F) = 3$ and $\Delta(F) = 1$, so it is excluded from the case $T \leq 1$. In every remaining case Chen, Wu and Xu supply a vertex v of degree at most 2. This vertex is not a cut vertex. It lies in at most one triangle, and $T(G - v) = T(G) - t(v) \leq 1$. The graph $G - v$ is connected, strong C_4-free and P_7-free, of order at least 5. Induction gives $\Delta(G - v) \geq 2$. Since $\Delta(G) = \Delta(G - v) + 2 - d(v) \geq \Delta(G - v)$, one has $\Delta(G) \geq 2$.
It remains to prove (4). Part (1) already gives the claim when $T \leq 1$. Suppose $T = 2$, and proceed by induction on n from order six. Lemma S5.6 supplies $\Delta \geq 0$ for every connected graph in the class of order at least two, and $\Delta \geq 1$ for every such graph of order at least four.
Suppose v is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma S5.5 gives $\Delta(G) = \sum_i \Delta(G_i) + (k - 1)$. A value at most 1 would require $k = 2$ and both sides of deficit 0, hence both of order at most three, and then $n \leq 5$. Thus $\Delta(G) \geq 2$.
Suppose a triangle is a clique cutset. Each side has order at least four, the side deficits add, and Lemma S5.6 gives $\Delta(G) \geq 2$.
Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. The side deficits again add. A side of deficit 0 has order three and three edges, so it is a triangle. Two such sides would contain a 4-cycle, so there is at most one. If $\Delta(G) \leq 1$, there are exactly two sides, one a triangle and the other a graph of order $n - 1 \geq 5$ with at most one triangle. Part (1) gives that side deficit at least two, a contradiction. This handles every clique cutset.
Suppose G has no clique cutset. The Petersen graph does not have exactly two triangles. The graph F of Lemma S5.6 has $T(F) = 3$, so it is excluded from the case $T = 2$. Theorem 1.1 of Chen, Wu and Xu, in the form already used for Lemma S5.6, therefore supplies a vertex x of degree at most two. The graph $J = G - x$ is connected, and $\Delta(G) = \Delta(J) + 2 - d(x)$. For $n \geq 7$ the inductive hypothesis gives $\Delta(J) \geq 2$ when $T(J) \leq 2$, while $T(J) \leq 1$ is the case of part (1); in both subcases $\Delta(G) \geq 2$. For $n = 6$ the same comparison leaves only the case $d(x) = 2$, $|V(J)| = 5$ and $\Delta(J) = 1$. Then J has six edges. A

hole in J would be a 5-cycle, and any chord of that cycle is a 4-cycle, so J is chordal. The simplicial deletion in the order-five argument above gives $e(J) = |V(J)| - 1 + T(J)$, hence $T(J) = 2$. Two triangles on five vertices cannot share an edge, for that graph contains a 4-cycle, and they cannot be disjoint, so they share exactly one vertex. Every pair of vertices of this graph has a common neighbour. Joining x to any two of them creates a 4-cycle, which contradicts $d(x) = 2$. □

Remark S5.8. Lemmas S5.6 and S5.7 are the deficit estimates that follow from Theorem 1.1 of Chen, Wu and Xu once clique cutsets of order 1, 2 and 3 are restored. The graph F of Lemma S5.6 has $\Delta(F) = 1$ and $T(F) = 3$ and lies outside the class, so only the Petersen graph is an exception inside the class. The resulting estimate is $\ell < 4$. Theorem 5.8 gives the sharper bound $S \le 3n$.

APPENDIX

A | The finite certificate

This appendix is the certificate cited in the proof of Lemma 5.7. Lemma 5.7 is proved by a finite search whose completeness argument is in the article. The completeness argument for Lemma S5.3 is in Section 5. This section records the audit. The decision arrays are not printed here, and they are not reconstructed in the text.

The program is submitted with the manuscript as a separate file. In the repository https://github.com/neardws/graph-conjecture-133 it is `certificates/gen3/verify_gen3.py` at commit `dd555d305b4a69076cafb845087f00b4144f07cc`. The SHA-256 digest of that file is `6be3852172aa401387b498c7e5a8a0efbbdde72aabb1e7fad0db9085b89eb86c`. The present revision does not alter the file. The program uses only the Python standard library, requires Python 3.10 or later, embeds the decision certificate, and takes no arguments. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`.

That command was executed independently for this revision, with the standard-library interpreter `python3`, from the repository root, and without modifying `verify_gen3.py`. The run printed the five branch rows and the passing line below. The first row is the one-triangle tree of Lemma 5.7. The other four rows are the initial configurations of Lemma S5.3, in the order shared vertex, one joining edge, and joining paths of lengths two and three.

```
one triangle           nodes=169  maximum order=17  leaves=52  unrestricted=4
triangle distance 0    nodes= 36  maximum order=17  leaves= 7  unrestricted=0
triangle distance 1    nodes= 31  maximum order=15  leaves= 6  unrestricted=0
triangle distance 2    nodes= 46  maximum order=11  leaves=21  unrestricted=0
triangle distance 3    nodes=  5  maximum order=10  leaves= 3  unrestricted=0
PASS. All branches and all decision entries verified.
```

The counts in the tables below are these five rows. No certificate array was rewritten for the run.

The induced-P_7 rejection is equivalent to a condition on ends of paths. Let S be a set of six vertices of the graph already built, and let z be the new vertex. Deleting one vertex from an induced P_7 leaves either a path on six vertices or two path components. In the first case the deleted vertex was an end, so it is adjacent to exactly one end of the remaining path. In the second case it was internal, so it is adjacent to exactly one end of each component and to no other vertex of S. An isolated vertex is a path of order one and is its own end. Conversely, if $G[S]$ is a disjoint union of one or two paths and the neighbourhood of z on S consists of exactly one endpoint of each path component and of no other vertex of S, then $S \cup \{z\}$ induces a P_7. A component that is not a path has either a vertex of degree greater than two or no end, and a set with any other number of edges is not one of these two graphs. The checker forbids precisely those neighbourhoods:

```
for each 6-set S:
    if some vertex of S has more than two neighbours in S: skip
    if the number of edges induced by S is not 4 or 5: skip
    split S into its components
    if a component has no end, or the number of components
        is not 6 minus the number of edges: skip
    reject each neighbourhood N for which N cap S consists
        of exactly one endpoint of each path component
        of H[S], and of no other vertex of S
```

The skip rejects a vertex of S only when it has more than two neighbours in S. The program named above implements that test by `max(degrees.values()) > 2`. The pseudocode is aligned with the pinned file, and the file is unchanged.

The checker labels each new vertex by the current order. A decision is an integer, a pair, or an unrestricted node. An integer names a vertex whose current deletion loss is less than 4, and every neighbourhood explored at that node contains the vertex. A pair (a, b) names two adjacent vertices of degree two. The neighbourhood U explored at that node is required

only to satisfy $U \cap \{a, b\} \neq \varnothing$. The verifier does not replace the pair by one predetermined endpoint: fixing a single endpoint in advance would drop every branch that meets only the other endpoint. The checker visits every legal child neighbourhood in increasing integer-mask order and consumes the supplied choices in preorder. A new 4-cycle exists exactly when two members of the new neighbourhood have a common neighbour in the graph already built. For a new induced P_7, the checker tests six old vertices together with the new vertex and reads their induced edges. At an unrestricted node it compares the labelled graph with $H_{3,2}$ and continues to explore the extensions. A missing choice, or an unused choice, makes the verification fail. The verifier imposes no order cutoff and performs no isomorphism pruning.

The same three tests are the audit of the run quoted above. A child is rejected when two vertices of the new neighbourhood have a common neighbour in the graph already built; that is the 4-cycle test, and it does not use an external list of cycles. A child is rejected when the new vertex and six existing vertices induce a P_7; the checker reads those induced edges and does not truncate the order. At an unrestricted node every nonempty neighbourhood is admitted, the labelled graph is compared with $H_{3,2}$, and the search continues. The one-triangle tree has four unrestricted nodes, all equal to $H_{3,2}$, and each of the four two-triangle trees has unrestricted count 0. A leaf is a node with no legal child. The expansion rule at a pair of adjacent degree-two vertices remains membership $U \cap \{a, b\} \neq \varnothing$, with neither endpoint fixed in advance.

The reason a finite tree covers every order is the one given in the proof of Lemma 5.7. If the current induced subgraph H has a vertex v with $L_H(v) < 4$ and v has no neighbour outside H, the degree and the triangle count of v are already final, which contradicts $L \geq 4$ in the finished graph. If a and b are adjacent vertices of degree two in H, at least one has a neighbour outside H. The new neighbourhood U is required to satisfy $U \cap \{a, b\} \neq \varnothing$. The condition is membership in the pair, not the prior selection of one endpoint. A deficient vertex v is the case $U \cap \{v\} \neq \varnothing$. When H already satisfies both local conditions, the search compares it with $H_{3,2}$. If H is a proper induced subgraph of a connected graph satisfying the hypotheses, some vertex outside H has a neighbour in H, and the node is unrestricted. For a fixed embedding into the target graph, the old neighbourhood of a required new vertex is one of the sets the node enumerates, and the child is kept. An embedded proper prefix therefore extends. Every potential counterexample continues along some branch.

Order	3	4	5	6	7	8	9	10
Visited nodes	1	1	1	1	2	3	5	14
Order	11	12	13	14	15	16	17	
Visited nodes	23	24	23	19	24	20	8	

The full one-triangle tree has 169 labelled nodes and 52 leaves. Four unrestricted nodes occur, all at order 11, and all four represent $H_{3,2}$. The search continues to expand each of those copies. The largest order reached is 17. That value is the height of the search tree: the verifier imposes no order cutoff.

The second certificate is the search in the proof of Lemma S5.3. It uses different initial graphs and different decisions. At each node the verifier checks a single vertex of current deletion loss less than 4 and explores every legal extension that meets that vertex. The four rows are the shared-vertex configuration, the single joining edge, and the joining paths of lengths two and three.

Initial configuration	Order	Nodes	Largest order	Leaves	Unrestricted
Shared vertex	5	36	17	7	0
One joining edge	6	31	15	6	0
Joining path of length two	7	46	11	21	0
Joining path of length three	8	5	10	3	0

These searches visit 118 nodes in all, with 37 leaves and no unrestricted node, and every branch closes. An induced embedding into a graph with $L \geq 4$ at every vertex would continue at every node, which is the local conclusion of Lemma S5.3.

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